

FoxPlus-A Connectors & Cables								
			CONNECTORS					
Conn / Cable	ISIS / AMSAT Supplied	Status	Manufacturer	Mfr Part No.	Connection	Male / Female	Interface	Notes
1	AMSAT	IH	Samtec	MTSW-106-08-L-D-235	PSA to ISIS CSKB 12 pin	Male	PSA J1, J3	
2	AMSAT	IH	Samtec	MTSW-107-08-L-D-235	PSA to ISIS CSKB 14 pin	Male	PSA J2, J4	
3	AMSAT	IH	Samtec	QSH-030-01-L-D-A	QSH Stacking Connector	Female	PSA J5	
4	AMSAT	IH	Harwin	M80-5301042	PSA to ISIS AntS (board)	Male	PSA J6	
5	AMSAT	IH	Harwin	M80-5301042	PSA to ISIS Solar Panels (board)	Male	PSA J7	
6	AMSAT	IH	Molex	5055651401	UMB Connector / plug – conn shell	Female	PIU J10 mate	This will be a plug for ABF. Do we need a cable here for testing? Female mating connector needs socket pins. These are crimp on and push in
7	AMSAT	IH	Molex	5054311000	UMB Connector / plug – crimp pin socket	Female	PIU J10 mate	Same as J9 socket pins.
8	AMSAT	IH			Seperation Switch (PIU board)		PIU J8 mate	
Not Used	ISIS	IH	Molex	505565-0681	CH3 I2C	Female	Not Used	Not used. Listed here for completeness. Same signals as on CSKB
9	AMSAT	IH	GCT	USB4235	External Umbilical Connector	Male	USB J1	New 8/23/26. USB-C connector mounts to antenna cover.
10	AMSAT	IH	Molex	5055670881	USB to PSA 8 pin	Female	USB J2	Edge connector on USB board
11	ISIS	IH	Omnetics	A28000-009	PSA to AntS	Female	AntS J1	ISIS provided Mating connector to A29100-009 male on AntS. Comes with
	CABLES							
USB_PSA	AMSAT	Needed	Molex	TBD	USB Board J2		Mates to Connector #10	This cable carries signals and power from the external UMB via USB board J2 to the PSA J8
	AMSAT	Needed	Harwin	M80-5301242	PSA J8			
UMBX	AMSAT	Needed	GCT	USB4235	External Umbilical Cable, sat side	Female	Connector #9	This cable goes from the external connector on the sat to the EGSE & Umbilical Attached switch
	AMSAT	Needed	Amphenol	DB25P064TXLF	External Umbilical Cable, EGSE side	Male	EGSE J5	DB25 with solder sockets
UMB1	AMSAT	Needed	Molex	505431100	PSA board to ISIS PIU	Female	PIU J9 mate	This cable goes from PIU J10 to the AMSAT connector #9. Same as J10 socket pins. UMBN combines UMB1 and UMB2
	AMSAT	Needed	Molex	5055651501	PSA board to ISIS PIU	Female	PIU J9 mate	Female mating connector needs socket pins. These are crimp on and push in. UMBN combines UMB1 and UMB2.
UMBT	AMSAT	TBD	Molex	5055651401	Test Cable (if needed)	Female	PIU J10 mate	Same as above J10 mate connector to incl socket pins. Don't know at this time if this cable is needed. TBD
	AMSAT	TBD	TBD	TBD	Test box connector	Female	TBD	
SP_PWR	ISIS	TST	ISIS?	A106268-001	Solar Panel Power, SP side	Female	Each Solar Panel	Power Connector on each of 5 solar panels
	AMSAT	TST	Molex	504051-0801	Solar Panel Power, PIU side	Female	PIU J11	3 J11 connectors: CH0, CH1, CH2 so 2 solar panels each
SP_TMP	ISIS	Needed	ISIS?	A105383-001	Solar Panel Temp Sens, SP side	Female		Temperature sensors on solar panels. Panel sequence TBD
	AMSAT	Needed	Harwin	M80-4611042	Solar Panel Temp Sens, PSA side	Female	PSA S1	
	AMSAT	Needed	Harwin		Solar Panel Temp Sens, crimp pins socket	Female		
SEP_SW	AMSAT	Needed	Molex	505565-0601	Seperation Switch I/F on PIU	Female	PIU J8 mate	
	AMSAT	Needed			Seperation Switches on Frame			Solder to switches on frame? TBD
SEP_TST	AMSAT	TBD	Molex	505565-0601	Seperation Switch Test (if needed)	Female	PIU J8 mate	J8 is the separation switch interface on the PIU
	AMSAT	TBD	TBD		Seperation Switch Test (if needed)		Ext Test box?	
BATT	ISIS	IH	Molex	505565-1501	Battery Connector, PIU side	Female	PIU J2 mate	This cable supplied by ISIS
	ISIS	IH			Battery Connector, PBP side			Battery Side is supplied connected?
ANT_IF	AMSAT	TST	Harwin	M80-5301042	Antenna Interface cable, PSA side	Female	Connector #4 Mate	PSA mating connector
	AMSAT	TST	Omnetics	A28000-009	Antenna Interface cable, Ant side	Female	Connector #11	ISIS Ant-S mating connector
ANT_VHF	AMSAT	Needed	AMSAT?		ICR to VHF Dipole Antenna	Female	AntS J2	MMCX Connector
ANT_UHF	AMSAT	Needed	AMSAT?		RXTX to UHF Dipole Antenna	Female	AntS J3	MMCX Connector
	CONNECTOR TOOLS							
	AMSAT	IH	Molex	2002180300	Hand crimp tool for Molex 1.25" Pitch			Available from Digikey
	AMSAT	IH	Molex	638270800	Hand crimp tool for Molex 1.50" Pitch			
	AMSAT	IH	Molex	638132700	Insertion / Extraction tool			
	AMSAT	IH	Harwin	M22520/2-01	DMC Hand crimp tool for Harwin M80			

NOTES: IH = In House
IH = In House / Complete
TST = Test Cable Complete